arXiv:2610.26390v1 [cs.RO] 19 Oct 2026

Cartender: Verifiable Language-Conditioned Driving by Learning the Drives Not Taken

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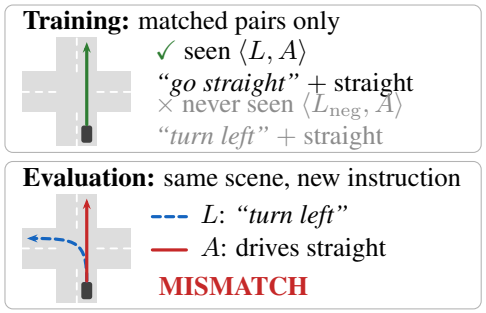
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Abstract

Every driving scene is logged exactly once, yet driving is inherently one-to-many. The scene decides which behaviors are admissible and the driver's intent which one to take: vision alone cannot choose among them, and language alone cannot tell what the scene permits. But the log keeps only the drive that was taken, and its instruction merely describes it. Language-conditioned driving policies are reported to follow instructions, yet on such data this cannot be established: a policy that ignores the instruction and one that obeys it regardless of the scene fit the logs equally well, and neither is distinguishable from one that weighs both. Trained only on such logs, driving vision–language–action models (VLAs) and world–action models (WAMs) receive no direct supervision on actions that violate their instructions or instructions the scene rules out. We present **Cartender**, which learns from the drives not taken. To construct matched and mismatched instruction–action pairs from real recordings without a simulator, we propose *Counterfactual Branching*, which treats each logged trajectory as one branch among several candidate futures: a vision–language model proposes the others and the instruction that selects each, and instructions the scene cannot support become negative labels. Cartender then weighs both explicitly: a learned scene–language compatibility regulates language conditioning during trajectory generation, and a reward-trained language residual re-ranks the candidates over a scene-based score. Branches serve training; instruction following is evaluated by changing instructions under identical recorded inputs, using behavior criteria rather than generated trajectory targets. On WOD-E2E, the best of four proposals reaches an RFS of 8.642 across 3,603 feasible instruction conditions. Across all 8,974 admitted proposals, mean RFS is 8.203 versus 7.906 for their matched logged trajectories, and 67% score higher than their logged counterpart. Relative to matched-only imitation with the same backbone, Cartender more than doubles the rate of fulfilling both instructions of a same-scene pair (57.00% vs. 23.67%) while RFS rises from 6.32 to 7.10; its gate cuts execution of infeasible instructions from 53.36% to 14.18%.

1 Introduction

The same driving situation admits many competent behaviors — overtake the slow truck now or follow it for three seconds first, take the gap at 40 km/h or at 25, return to the right lane the moment it is clear or stay out until the exit — and the scene settles only which are admissible, never which one is preferred (Liao et al., 2025; Wang et al., 2026b). That is settled by an intention the sensors never see. They record the situation, not the reason one behavior was chosen, and not the behaviors passed over. Nor can more data recover the



rest. A vehicle occupies one position in space and time
and drives one trajectory; the situation never reassembles itself for a second attempt. Every driving corpus therefore contains single samples from genuinely multi-modal conditional behavior distributions — one sample each, whatever else is recorded alongside it. The alternatives are not missing from the datasets we happen to have. They are not collectible as outcomes of that same event.

Their absence is felt twice: once in what can be trained and, with far less attention, once in what can be checked. Matched records show the executed behavior, leaving alternative responses unobserved. Establishing instruction control requires testing what the policy does under a *different* instruction in the *same* scene. Matched instruction–trajectory evaluation leaves this response untested. The instruction-following scores reported on such data are taken to show that the policy reads the instruction. They do not. The visual scene may not determine which admissible behavior is preferred, but it can already narrow the space of plausible actions enough for a largely scene-driven policy to fit the recorded trajectory well; an instruction-sensitive policy can fit the same matched pair just as well, while differing only in what happens when the instruction is changed. Both score well. Figure 1 makes the ambiguity concrete. Only an instruction intervention separates them.

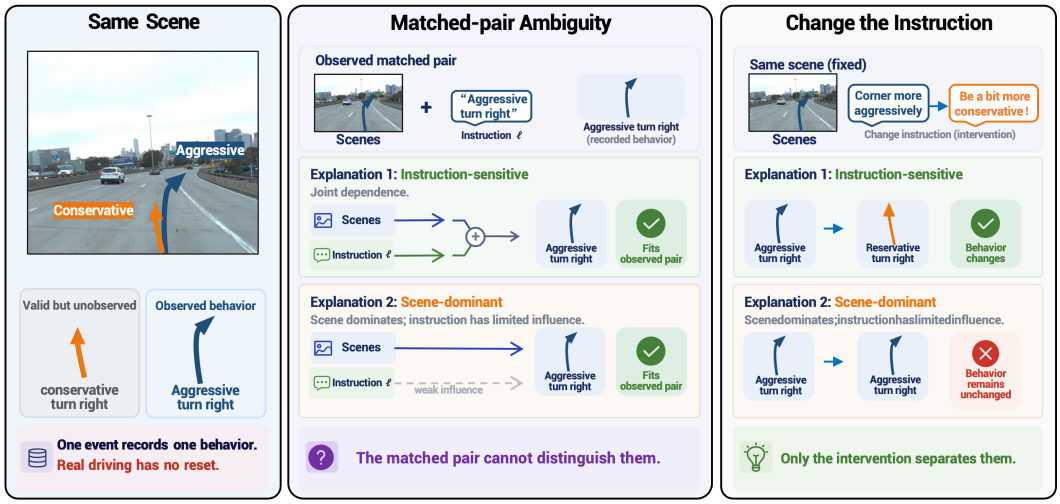


Figure 1: **Matched-pair ambiguity in real driving.** A single event reveals one recorded behavior, so the same matched instruction–trajectory pair can be consistent with policies exhibiting different instruction dependence. Holding the scene fixed and changing the instruction separates them.

Simulation can construct alternative trajectories from a recorded simulator state. SimLingo's Action Dreaming pairs language instructions with alternative trajectories generated from recorded CARLA states through offline, non-reactive simulation (Renz et al., 2025). Our construction instead operates on real recordings without a simulator. Matched-pair evaluation on these recordings retains the ambiguity above. Whether same-scene supervision helps is not the open question; simulation already indicates that it does. Two others are. The first is about data: whether such supervision can be obtained, and trusted, where the situation cannot be replayed. The second does not wait on the first. Instruction dependence can already be measured on released models, and that is where we begin.

Two of the three diagnostics we use to measure instruction dependence need no data beyond what a model was already trained on. Instructions are first shuffled across scenes. If the action loss does not move, the matched objective is insensitive to instruction identity. A rise exposes dependence. At inference, removing the instruction tests whether its presence changes the predicted trajectory. Neither diagnostic requires a behavior that was never driven, so the confound becomes something to measure rather than something to argue about. On LMDrive, shuffling increases waypoint L1 loss by 0.011. Under the evaluated Qwen-

Drive interface, instruction removal shifts predictions by only 0.064 m. The third holds the scene fixed and changes the instruction, probing alternative behaviors that a single recorded event cannot jointly reveal.

We construct same-scene, multi-intent supervision on real driving data, treating its two halves differently. Declaring an instruction *inapplicable* to a recorded scene takes a label and no trajectory at all, so scene–instruction applicability is grounded in real observation by construction; supplying the behavior a *feasible* alternative instruction would have produced takes a trajectory that was never driven, and Sec. 3.1 states how it is branched and on what grounds it is admitted. We then keep training and measurement apart. Branched alternatives enter training. Evaluation changes instructions under an identical recorded input, with both requests reviewed for feasibility and predictions assessed by behavior criteria, not by distance to generated teachers.

The feasible half is the hard one: it requires the behavior that a valid alternative instruction would have produced, and we supply it with *Counterfactual Branching*. At selected keyframes, a vision–language model proposes route-relative parameters and corresponding instructions; deterministic construction and rule-based admission produce trajectory branches in the planner's prediction space. Branch quality is an empirical question, not a design assumption. We evaluate admitted branches against matched logged trajectories using Rater Feedback Score (RFS). Across the complete set of 3,603 feasible instruction conditions, the branched set reaches a best-of-four RFS of 8.642. Across all 8,974 admitted proposals, mean RFS is 8.203, compared with 7.906 for their matched logged trajectories, and 67% of branches individually outperform their logged counterparts.

A demonstration identifies a point; a preference among admissible behaviors is an ordering. That distinction follows from multimodality: a demonstration says "drive like this," but does not rank admissible alternatives under an instruction; a reward expresses that ordering. The supervision supplies the candidates, not the ordering. Same-scene alternatives make the comparison expressible; the ordering itself is defined by the frozen key-event reward in Sec. 3.2. Cartender accordingly generates a candidate set, estimates continuous scene–language compatibility to scale language conditioning during generation and ranking, and ranks the candidates with a demonstration-initialized base score plus a reward-trained language residual. Our main contributions are:

1. **The matched-pair confound, and how to measure it.** An instruction-following score on matched instruction–trajectory pairs does not establish that the policy's behavior depends on the instruction. We expose this ambiguity through instruction removal, shuffling, and same-scene swapping. Removal and shuffling, where native action loss is accessible, need no alternative trajectory targets; swapping fixes a recorded input and tests two feasible requests without generated trajectory targets.

2. **Counterfactual Branching.** Same-scene, multi-intent supervision built on real recordings without a simulator, where the logged trajectory anchors route-relative candidate futures, each paired with a proposed instruction. Branches pass rule-based admission before RFS evaluation: admitted branches average 8.203 against 7.906 for matched logged trajectories, with 67% individually scoring higher. Inapplicable instructions require labels but no synthesized trajectories.

3. **Cartender.** A policy that gates language conditioning through scene–language compatibility and ranks candidates with a reward-trained language residual over a demonstration-initialized base score. Cartender more than doubles maneuver paired success over matched-only imitation (57.00% vs. 23.67%), with RFS rising from 6.32 to 7.10. Gating reduces unsupported-request following from 53.36% to 14.18%.

2 Related Work

Multi-Proposal Trajectory Planning. Multi-proposal planners address behavioral multimodality by predicting several trajectories per scene. DiffusionDrive (Liao et al., 2025) and DiffusionDriveV2 (Zou et al., 2025) generate candidates through truncated diffusion and select among them with a learned scorer. The candidate set hedges against missing one answer; it does not establish which answer the instruction prefers. We supply what matched demonstrations alone do not—a same-scene criterion for intention-dependent selection.

Driving Vision–Language–Action Models. LMDrive (Shao et al., 2024a), ORION (Fu et al., 2025), AutoVLA (Zhou et al., 2025), and MindVLA-U1 (Huang et al., 2026) connect language with trajectory generation through different architectures. NudgeVAD (Yang et al., 2026) adds language-conditioned residuals to a frozen planner and examines when navigation commands make language redundant. The central issue here is not architecture but what matched-pair training and evaluation establish. On matched pairs, the instruction and recorded behavior co-occur; agreement with that behavior alone does not establish that the prediction changes when the instruction does. ICR-Drive (Hamid et al., 2026) tests robustness to instruction perturbations under matched simulator routes and seeds. Our question is whether different feasible instructions select different appropriate behaviors under the same recorded input. Our contribution is diagnostic before it is architectural: we distinguish instruction dependence from agreement with a matched trajectory, with two tests requiring no new annotation.

Language–Action Alignment and Driving Supervision. LinkVLA (Wang et al., 2026a) strengthens language–action coupling through shared representations, while Drive My Way (Wang et al., 2026b) learns driving preferences. SimLingo's Action Dreaming (Renz et al., 2025) constructs alternatives through offline, non-reactive simulation; MALT-Drive (Lu et al., 2026) uses rollout-based counterfactual supervision to align control instructions with planning. Our setting requires alternative trajectories for real recordings without simulator rollouts. OmniDrive (Wang et al., 2025) supplies counterfactual descriptions of alternative outcomes; constructing the trajectories themselves additionally requires checking their validity. Counterfactual Branching does not sidestep this requirement. It constructs alternatives from recorded routes, admits them through rule-based checks, and evaluates their quality afterward using RFS (Sec. 3.1); these checks do not certify full scene feasibility. Generated teachers serve training, not evaluation targets: instruction control is assessed by behavior criteria under fixed recorded inputs.

3 Method

Given a visual scene s and an instruction ℓ, Cartender predicts an ego trajectory $\tau \in \mathbb{R}^{T\times 2}$. Same-scene relations supply two signals: which requests the scene supports, and which trajectories realize them. Figure 2 shows how these signals govern generation and candidate ranking.

3.1 Same-Scene Relational Supervision

A single visual scene can support multiple distinct driving behaviors while leaving other requested behaviors unsupported. We organize supervision around this shared context: feasible instructions receive teacher trajectories, and infeasible instructions receive negative applicability labels. The two halves are not equally hard to obtain. An infeasible relation needs no counterfactual trajectory; a feasible alternative requires behavior absent from the recording. Counterfactual Branching supplies it. At selected keyframes, a vision–language model proposes alternative futures and their instructions alongside the logged branch. We

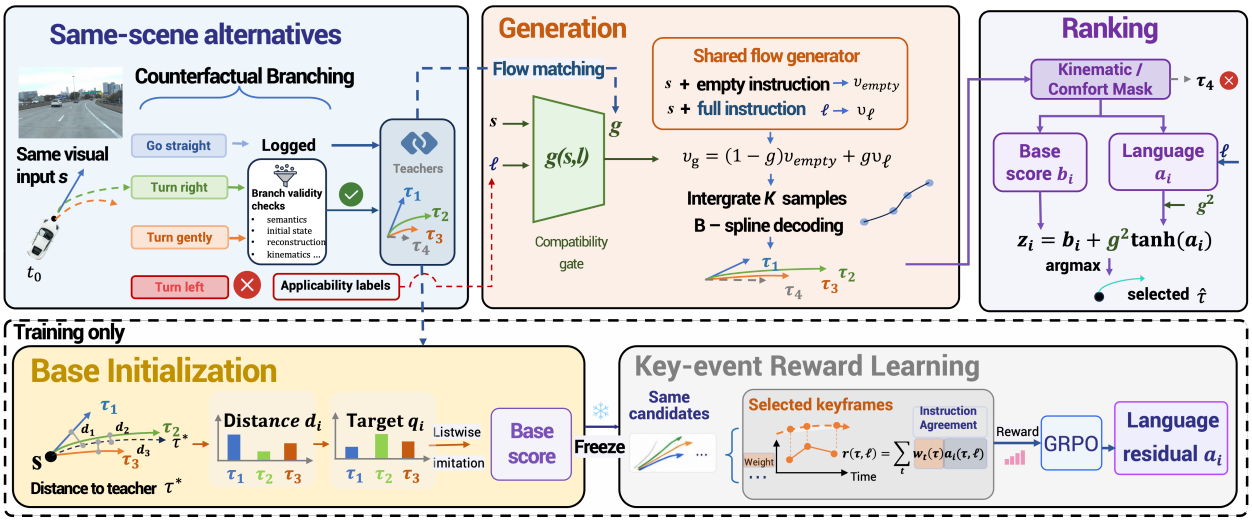


Figure 2: Overview of Cartender. Offline same-scene relations provide applicability labels for compatibility learning and teacher trajectories for flow matching and base-score initialization. Compatibility modulates generation and a language-dependent ranking residual. After base initialization, reward learning updates only the residual. At inference, the policy ranks candidates passing kinematic and comfort checks without teachers or reward evaluation.

reconstruct proposals in the planner's continuous trajectory space, enforcing observed initial position and velocity before boundary-preserving smoothing. We check source validity, semantic conflicts, initial-state consistency, reconstruction error, and kinematic and comfort violations, including those after reconstruction. Passing these checks does not establish driving quality. Section 4.4 evaluates the best branches and admitted set using automated Rater Feedback Score (RFS); RFS does not determine admission.

Branched trajectories supervise training. The diagnostic in Sec. 4.3 reuses one recorded input snapshot for two instructions independently reviewed as feasible. It does not pair approximately similar scenes or assume both responses were recorded. Behavior criteria assess predictions without generated trajectory targets.

Binary applicability labels train a continuous compatibility score. Let $q_\ell(k \mid \ell)$ be the frozen language-intent posterior and $p_s(k \mid s)$ the scene-support probability for intent k. The requested intent $\hat{k}$ and confidence c_ℓ define

$$\hat{k} = \arg\max_k q_\ell(k \mid \ell), \qquad c_\ell = \max_k q_\ell(k \mid \ell), \qquad g(s,\ell) = c_\ell \, p_s(\hat{k} \mid s). \tag{1}$$

We train support with intent-balanced binary cross-entropy on feasible and infeasible relations, then freeze the compatibility model. The language-intent branch remains fixed. The gate estimates maneuver-level support; the generator and ranking residual condition on the full instruction, including timing, speed, and manner. Section 4.3 evaluates maneuver and qualifier control separately.

3.2 Learning from Structured Language–Behavior Relations

Teacher trajectories supervise conditional flow matching in B-spline coefficient space; a fixed decoder D_{BS} maps coefficients c to T-step trajectories (Han et al., 2026). At flow time t and state c_t, the same generator produces empty-instruction and full-instruction fields. Compatibility sets their classifier-free guidance weight:

$$v_g(c_t, t) = v_\emptyset(c_t, t; s) + g(s,\ell)\,[v_\ell(c_t, t; s, \ell) - v_\emptyset(c_t, t; s)]\,. \tag{2}$$

With $g \in [0, 1]$, scene–language compatibility interpolates between the scene-conditioned empty-instruction field and the full-instruction field, controlling the strength of language conditioning. Integrating from K noise samples and decoding yields $\mathcal{T}(s, \ell) = \{\tau_i\}_{i=1}^{K}$.

We mask candidates violating kinematic or comfort constraints and denote the surviving indices by $\mathcal{A}$. Their logits combine an imitation-initialized base with a bounded language residual:

$$z_i = b_\psi(s, \tau_i) + g(s, \ell)^2 \tanh a_\theta(s, \ell, \tau_i). \tag{3}$$

The base reads the scene and trajectory; the residual also reads the instruction. Scaling by g^2 limits language-driven ranking changes at low compatibility. Inference selects the highest-scoring candidate under a masked softmax; if none survives, it returns no trajectory.

Geometric proximity to a teacher need not imply instruction fulfillment. We therefore initialize ranking by teacher proximity and refine it with the key-event reward. For each feasible relation (s, ℓ), proximity to its admitted A and B teachers $\mathcal{H}(s, \ell)$ defines a target distribution q_i over $i \in \mathcal{A}$. C and D are not teacher targets. We train the base scorer with

$$\mathcal{L}_{\mathrm{IL}} = -\sum_{i \in \mathcal{A}} q_i \log \pi_\psi(i \mid s, \mathcal{T}), \tag{4}$$

averaged over feasible relations, where π_ψ is the masked softmax of base scores. During base initialization, the language residual stays fixed at zero.

We then freeze the generator, compatibility model, and base scorer. Only the language residual learns, through Group Relative Policy Optimization (GRPO) (Shao et al., 2024b). A frozen key-event evaluator supplies the reward

$$r(\tau, \ell) = \sum_{t=1}^{T} w_t(\tau)\, a_t(\tau, \ell), \qquad \sum_{t=1}^{T} w_t(\tau) = 1, \tag{5}$$

where t indexes trajectory steps, w_t emphasizes salient motion changes, and a_t measures trajectory-observable instruction agreement. GRPO normalizes relative advantages within each scene–instruction group before weighting reward-driven updates by g^2. Low-compatibility groups receive a stronger anchor to the frozen imitation policy.

Finite candidates and frozen rewards also permit exact summation of the selector's expected reward over unmasked candidates. GRPO is therefore an optimization choice rather than a requirement of the formulation. Section 4.5 compares both optimizers and tests gating, preference rewards, and staged optimization. Teachers and reward evaluation are training-only. Low compatibility attenuates language influence in generation and ranking; it does not produce an explicit refusal.

4 Experiments

4.1 Experimental Setup

Data and comparisons. Our data pool comprises 10,336 training and 4,580 validation scenes from WOD-E2E (Xu et al., 2025); the quality evaluation uses 1,200 validation scenes. Unless otherwise stated, Cartender variants and the language-free control share frozen U1 scene features, a 5 s horizon, and $K = 24$ candidates. Released-policy diagnostics use fixed checkpoints of Qwen-Drive (Zhou et al., 2026), SimLingo (Renz et al., 2025), and LMDrive (Shao et al., 2024a): Qwen-Drive uses our added instruction interface on WOD-E2E; SimLingo and LMDrive use their respective CARLA protocols.

Table 1: Instruction dependence and same-scene control.

(a) Released-policy diagnostics.

Model	Ref. ADE (m)	Removal shift (m)	Shuffle $\Delta\mathcal{L}_{\text{action}}$	Swap paired success (%)
Qwen-Drive	1.413	0.064	—	21.13
SimLingo	0.842	2.511	+0.232 +1.289	50.00
LMDrive	0.203	0.178	+0.011	3.20

(b) Same-scene instruction control.

Model	Maneuver success (%)	Trajectory shift (m)	Direction consistency (%)
Vision-only	20.37	0.00	0.00
Qwen-Drive	27.78	0.60	22.22
SimLingo	72.80	0.89	62.50
Cartender	**73.83** **± 2.60**	1.666 ±0.09	**83.09** **± 7.40**

(a) Frozen checkpoints follow their protocols. Ref. ADE uses original instructions; Removal uses empty inputs. Shuffle reports shuffled-minus-original native action loss, with scenes and targets fixed. SimLingo: route loss above speed-waypoint loss; LMDrive: waypoint L1. Dash: unavailable; shading: within-model contrasts. Paired success requires both requests fulfilled.
(b) Scenes and random draws are fixed. Maneuver success measures individual request fulfillment; trajectory shift measures prediction displacement; direction consistency measures alignment with the requested direction.

Supervision and training. Counterfactual Branching constructs 3,603 feasible instruction conditions across 2,487 source scenes. Four proposals per condition yield 14,412 trajectories, of which 8,974 pass rule-based admission. The admitted A and B proposals form the teacher sets; all 3,603 A proposals pass admission, while C and D are excluded from teacher supervision. Infeasible instructions supply applicability labels only. Selector comparisons use three seeds with a shared generator checkpoint, candidate sets, and fixed reward (Sec. 3.2); Table 3 reports mean paired success.

Evaluation. We test instruction dependence through removal, shuffling where native action loss is accessible, and same-scene swapping between feasible instructions. Maneuver-level and qualifier-level results are reported separately. The maneuver diagnostic uses 200 pairs; the qualifier diagnostic uses 64 speed pairs and 32 timing pairs. Paired success requires both requests to satisfy action and hard-constraint checks. Behavioral criteria are independent of Cartender's relational training annotations. We also report RFS, ADE, and output coverage. Missing predictions count as failures in success metrics; quality statistics state their coverage.

4.2 Is the Instruction Being Read at All?

Table 1a tests frozen released models under their respective input interfaces and datasets, without our constructed supervision. These are within-model interventions, not a common cross-model benchmark; Qwen-Drive results are specific to the added instruction interface. We evaluate instruction dependence using three controlled interventions: instruction removal, instruction shuffling, and same-scene instruction swapping.

Table 1a exposes the gap between predicting the recorded trajectory and following the requested behavior. Under the evaluated Qwen-Drive interface, instruction removal changes predictions by only 0.064 m, while same-scene paired success is 21.13%. LMDrive attains a reference ADE of 0.203 m but only 3.20% paired success under its evaluated protocol. These results show that low matched trajectory error does not establish reliable instruction control. SimLingo clearly uses the instruction: removal shifts predictions by 2.511 m, and shuffling increases both native route and speed-waypoint losses. Nevertheless, paired success is 50.00%, leaving at least one request unfulfilled in half of the pairs. Using language does not ensure

Table 2: Counterfactual branch quality and validation. RFS is evaluated after rule-based admission. Best-of-four averages conditions; mean admitted RFS and "Above logged" pool admitted proposals. "Above logged" is the fraction exceeding matched logged RFS. Two reviewers audit constructed infeasible relations.

Branch quality	
Matched logged RFS	7.906
MindVLA-U1 ref. RFS	7.92
Best-of-4 RFS	8.642
Mean admitted RFS	8.203
Above logged (%)	67.0

Construction set	
Source scenes	2,487
Feasible conditions	3,603
Generated proposals	14,412
Admitted proposals	8,974
Admission rate (%)	62.27

Negative-relation audit	
Audited relations	228
Confirmed infeasible	202
Confirmation rate (%)	88.6

that the requested behavior is realized; instruction dependence and behavioral fulfillment must be evaluated separately.

4.3 Instruction Dependence on Real Same-Scene Pairs

The diagnostic reuses recorded input snapshots; generated teachers are not evaluation targets. We fix observations and sampling noise while comparing two instructions feasible under the same observation. Table 1b shows that the vision-only control fulfills 20.37% of requests without changing its output; fulfillment alone does not establish instruction control. Cartender achieves 83.09% directional consistency. The response extends to execution intensity within the same maneuver. For gradual versus strong acceleration or deceleration, speed-direction accuracy is 64.58% versus 43.75%; requiring the correctly directed speed difference to exceed $0.2\,\mathrm{m/s}$ still yields 53.13% success versus 12.50%. Timing control remains limited, with 35.42% correct early–late ordering across 32 deceleration pairs.

4.4 Counterfactual Branching: Branch Quality

We evaluate constructed branches using RFS, which does not determine admission. Best-of-four RFS measures the best available candidate among A–D for each feasible instruction condition. Mean admitted RFS and the fraction exceeding matched logged RFS summarize all admitted proposals, not only the A/B teachers used for training.

Table 2 reports two aggregation units. Best-of-four takes the maximum among A–D for each of 3,603 feasible conditions before averaging. Mean admitted RFS and "Above logged" pool all 8,974 rule-admitted proposals, with each proposal compared against its condition's logged trajectory. Under this proposal-level aggregation, admitted branches average 8.203 RFS, exceeding the matched logged score of 7.906 by 0.297. Moreover, 67% score above their corresponding logged trajectories. The RFS advantage over matched logged trajectories therefore extends to a majority of admitted branches. These statistics use the complete construction across 2,487 source scenes rather than a sampled subset. A separate audit has two reviewers assess each of 228 constructed infeasible relations independently. Cases with different judgments receive an additional review. The audit confirms 202 relations (88.6%); 26 remain unconfirmed.

4.5 Main Results and Elimination of Alternative Explanations

We first evaluate whether instruction control can be improved while preserving trajectory quality. Beyond this evaluation, we analyze whether the gains can be attributed to additional same-scene examples, training schedules, reward optimization choices, or specific supervision designs.

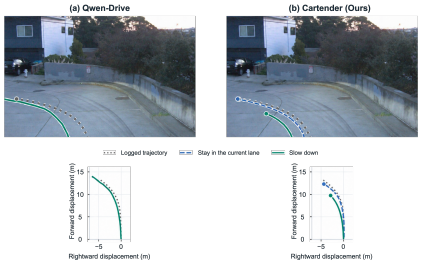
Instruction control and trajectory quality. Table 3 shows that Cartender more than doubles maneuver paired success over matched-only imitation with the same backbone, from 23.67% to 57.00%, while RFS rises from 6.32 to 7.10. Against the language-free control, maneuver and qualifier paired success rise from zero to 57.00% and 35.07%, respectively. RFS rises from 6.24 to 7.10, while output coverage falls from 98.98% to 97.70%. Paired success requires both requests to be fulfilled; missing outputs count as failures.

Contribution beyond additional trajectory examples. Table 3 compares matched-only and same-scene imitation under the same imitation procedure and optimization budget. Adding admitted A/B teachers raises maneuver paired success from 23.67% to 39.50%, a gain of 15.83 percentage points. On the same expanded training set, Cartender further improves maneuver and qualifier paired success by 17.50 and 26.04 points, reaching 57.00% and 35.07%. Its RFS is 7.10 compared with 7.04 for same-scene imitation, while coverage increases from 94.78% to 97.70%. The full training procedure therefore improves instruction fulfillment beyond imitation on the additional examples.

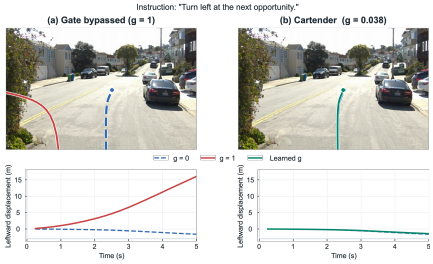
Compatibility and preference reward. Table 3 shows that adding compatibility to same-scene imitation raises maneuver and qualifier paired success from 39.50% and 9.03% to 45.83% and 12.85%, respectively, while RFS decreases from 7.04 to 6.95. The full method reaches 57.00% and 35.07%, with RFS increasing to 7.10. Removing compatibility from the full method reduces maneuver paired success by 15.33 percentage points to 41.67%, eliminates qualifier paired success, and lowers RFS from 7.10 to 7.00 and coverage from 97.70% to 94.40%. Replacing reward refinement with language-conditioned residual imitation, using the same initialization and update count, reduces maneuver and qualifier paired success by 7.00 and 30.56 points, respectively, and lowers RFS from 7.10 to 6.37. Both components therefore contribute to instruction fulfillment, with compatibility also supporting output availability.

Staged versus joint optimization. Both variants start from the same IL endpoint; the joint variant additionally retains the imitation loss during preference refinement. Table 3 shows that joint optimization and staged optimization both achieve 57.00% maneuver paired success. Staged optimization achieves 35.07% qualifier paired success versus 5.21% for joint optimization, and RFS of 7.10 versus 6.94. Staged optimization therefore favors qualifier fulfillment and trajectory quality without changing maneuver paired success in the reported runs.

GRPO versus exact expected-reward optimization. We compare GRPO with exact expected-reward optimization over the same finite candidate set, using the same reward and imitation anchor. Table 3 shows similar maneuver paired success: 57.00% for GRPO and 58.17% for exact expectation. GRPO achieves 35.07% qualifier paired success, compared with 4.86% for exact expectation, and RFS of 7.10 versus 7.00. This fixed-budget result favors GRPO on qualifier fulfillment, but does not isolate a general optimizer advantage.



(a) Same-scene instruction intervention.



(b) Compatibility gating under an unsupported instruction.

Figure 3: Cartender responds to a supported slow-down request under fixed scene inputs (left) and suppresses an unsupported turn through compatibility gating (right).

Table 3: Instruction control, trajectory quality, and output coverage. All methods use logged data and admitted A/B teachers except matched-only imitation. Imitation baselines share procedure and budget; exact expectation shares GRPO's candidates, reward, and imitation anchor. Paired success averages three seeds and requires both requests to succeed; missing outputs fail. Coverage measures output availability.

Method	Maneuver paired (%) ↑	Qualifier paired (%) ↑	RFS ↑	Coverage (%) ↑
Language-free control	0.00	0.00	6.24	98.98
Matched-only imitation	23.67	3.13	6.32	96.40
Same-scene imitation	39.50	9.03	7.04	94.78
Same-scene imitation + compatibility	45.83	12.85	6.95	97.70
Cartender	57.00	35.07	7.10	97.70
Cartender, no compatibility	41.67	0.00	7.00	94.40
Cartender, no preference reward	50.00	4.51	6.37	97.70
Joint IL + preference	57.00	5.21	6.94	97.70
Exact expected reward	58.17	4.86	7.00	97.70

Qualitative instruction interventions. In Fig. 3a, Qwen-Drive predictions nearly overlap for lane keeping and slowing down, whereas Cartender predicts a shorter trajectory for slowing down. In Fig. 3b, bypassing the gate produces pronounced leftward motion, while the learned gate keeps the prediction close to the $g = 0$ reference when the turn is unsupported within the evaluated horizon.

4.6 Behavior Under Infeasible Instructions

We test whether scene–language compatibility distinguishes unsupported instructions and reduces their influence on predicted behavior.

Compatibility separation. Mean compatibility scores are 0.771 for feasible instructions and 0.310 for infeasible instructions, with overlapping distributions. The score achieves an AUROC of 0.863 for identifying infeasible instructions. At $\theta = 0.5$, classifying low-compatibility instructions as infeasible yields 76.74% precision, 74.16% recall, and 75.43% F1.

Behavioral suppression of unsupported instructions. With checkpoint, inputs, and random draws fixed, learned compatibility reduces infeasible execution from 53.36% with $g = 1$ to 14.18%, and drift from 3.612 to 0.337 m. A separate request-level diagnostic defines Feasible Follow as valid, hard-constraint-passing fulfillment of feasible requests, and Relation Handling as its equal-weight average with valid, hard-constraint-passing nonexecution on infeasible requests. From no gate to learned compatibility, Feasible Follow falls from 61.6% to 43.8%, while Relation Handling rises from 54.1% to 64.8% (61.9% with a constant gate). These request-level metrics differ from paired success. Gating therefore trades feasible-request fulfillment for reduced unsupported execution in that diagnostic; it provides no explicit refusal.

5 Conclusion

Logged driving records one behavior even when several are admissible, so matched pairs provide neither a direct comparison among feasible behaviors nor evidence that language caused the selected one. Counterfactual Branching constructs same-scene alternatives with rule-based admission and post-hoc RFS evaluation.

These relations supervise training, while fixed-scene interventions on recorded inputs use behavior criteria to test whether predictions follow the changed instruction.

Cartender generates multiple candidates, gates language conditioning by continuous scene–language compatibility, and ranks them with a GRPO-trained language residual. It achieves 57.00% maneuver paired success versus 23.67% for matched-only imitation; gating reduces unsupported-request following from 53.36% to 14.18%. These results use fixed recorded inputs: rule-based admission and RFS do not certify collision freedom or instruction fulfillment, and the gate does not eliminate unsupported execution. Future work will test continuous instruction updates in closed-loop, interactive traffic.

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